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Education

SRM Institute of Science and Technology

Chennai, India

B.Tech, Computer Science and Engineering | CGPA: 9.4/10

June 2023 – May 2027

Technical Skills

Languages:	Python, SQL, JavaScript, Java, C++
ML & Data Science:	scikit-learn, TensorFlow, XGBoost, pandas, NumPy, OpenCV, feature engineering, time-series modelling, predictive analytics, model evaluation
Data & Backend:	PostgreSQL, PostGIS, MySQL, Flask, FastAPI, Node.js/Express, REST APIs, schema design, query optimisation, ETL and Excel/CSV pipelines
AI Engineering:	LLM API integration (Gemini, GPT-4, Claude), prompt engineering, retrieval-based verification
Tools & Web:	React, Next.js, Three.js, Docker, Git, Linux, Jupyter, pytest, Blender API (ezdxf)

Experience

Shimi Designs and Consulting Engineers Pvt Ltd

Chennai, India

Software Developer

August – September 2024

- Built a Flask and MySQL inspection-checklist web application that replaced a paper-and-verbal quality-assurance process used by roughly 15 engineering staff, reducing per-drawing verification turnaround from days to minutes.
- Digitised 91 structural-steel detailing checkpoints across five inspection categories, backed by a three-table normalised MySQL schema with cascading foreign keys and fully parameterised queries.
- Implemented session-based authentication with admin/user role separation enforced at both route and template level, plus one-click XlsxWriter export of per-product inspection records to formatted Excel workbooks.

STRCO India Pvt Ltd

Erode, India

Software Development Intern

June 2025

- Delivered a member-based business community website with a three-person team inside a four-week internship, covering public landing pages, member registration and profiles, role-gated content access, a member directory, and form-based inquiry capture.

Projects

ArchX3D — 2D CAD to Interactive 3D Building Pipeline | *Python, Blender, FastAPI, Three.js, Next.js* 2025 – Present

- Built a 56,000-line Python pipeline converting 2D DXF floor plans into furnished, textured 3D buildings, driving headless Blender for geometry extrusion, material assignment, and lighting.
- Wrote a furniture placement solver using candidate generation and scoring rather than constraint solving, ranking poses on wall affinity, door-swing clearance, window obstruction, and preservation of walkable circulation; items are placed in importance order and become obstacles for later ones, keeping layouts deterministic and explicable.
- Built an interactive browser viewer with orbit and first-person walk modes, using a merged BVH and capsule solver for collision with sliding and sub-stepping, a five-tier roof detection ladder from GLB metadata down to geometric inference, six visibility modes, and room fly-to navigation with an SVG minimap.
- Cut the viewer route's first-load JavaScript from 461 kB to 109 kB by deferring the 3D stack behind a client-only dynamic import, and kept all movement and classification logic free of rendering dependencies so 117 unit tests run in under 200 ms without a GPU.

Predictive Network Traffic Management | *FastAPI, TensorFlow, PostgreSQL, Docker, Next.js*

2025

- Built a software-defined networking traffic system benchmarking Random Forest, XGBoost, LSTM, and GRU models for congestion forecasting, served through a FastAPI backend with a real-time monitoring dashboard.
- Engineered time-series features including cyclical hour encoding, rolling mean/standard-deviation/max statistics across 5, 10, and 30-step windows, and multi-step lag features to capture temporal traffic patterns.
- Containerised the full stack — API, PostgreSQL 16, and SDN controller — with Docker Compose for reproducible deployment.

Smart City Civic Issue Tracker [Live Demo] | *React, Node.js, PostgreSQL, PostGIS*

2025

- Built a full-stack civic reporting platform with PostGIS-backed spatial storage and retrieval of geotagged issue reports using GeoJSON and WKT.
- Implemented JWT authentication with bcrypt hashing and three-tier role-based access control (citizen, officer, administrator) enforced across both API routes and React components.
- Integrated browser geolocation and camera capture with an OpenStreetMap Nominatim geocoding fallback for blocked GPS, plus optimistic UI updates for latency-free interaction.

Lake Maintenance Cost Prediction | *Python, scikit-learn, Streamlit*

2025

- Built a Random Forest regression model estimating irrigation lake maintenance cost from dimensional, inflow, and prior-expenditure features, deriving condition indices from current-versus-target depth gaps and pairing them with rule-based maintenance scheduling in a Streamlit interface.

Achievements

- **TechHacks National Hackathon** — 2nd Runner-Up among 100+ teams, March 2025.
- **Hack the Horizon, VIT** — built an LLM claim-verification system with OCR and speech-to-text input, cross-checked against Wikipedia and news sources, August 2025.
- **LeetCode** — 136 problems solved across data structures and algorithms.